

Evaluation Report

PII Redaction Tool - evaluation approach and results | Arun

Document: Red Herring Prospectus, KSH International Limited, dated December 10, 2025 (519 pages) **This run:** rules only, no spaCy model available while I was building it (see README) **To reproduce:** `python evaluate.py --source "Red Herring Prospectus.docx" --unit-tests`

1. How I evaluated it

Why I score per token instead of per span

I started out scoring whole spans and threw it away, for two reasons.

First, it's unfairly harsh on the free-text types. My detector finds Kirtane & Pandit inside the gold span Kirtane & Pandit, LLP. That's nearly all of the useful work done, but exact span matching calls it one false positive *and* one false negative, which tells me nothing about whether the tool is any good.

Second, accuracy needs true negatives to mean anything, and there's no sensible unit for "a span that was correctly not detected". Per token, every ordinary word is a real true negative, so accuracy is actually well defined.

So each token gets one gold label and one predicted label - a PII type, or 0 - and precision, recall and F1 come out of the confusion matrix. Accuracy is just correct tokens over all tokens.

The gold standard

I couldn't hand-annotate 5,041 paragraphs, so I took a stratified sample of 140 paragraphs (6,968 tokens) and labelled every PII mention in each one:

Stratum	Paragraphs	Tokens	Gold PII tokens	Why
A - PII-dense	70	3,785	261	Measures recall where the PII actually is
B - random prose	70	3,183	90	Measures false positives on text that's nearly all negatives

Stratum A was drawn from paragraphs matching contact-block and governance cues (Contact Person, Registered Office, Telephone, Promoter, Director, DIN, @). Stratum B was drawn uniformly from everything else. Sampling is seeded so it's reproducible.

The detector is fitted on the whole source document, not on the sample. Fitting on the sample alone would have flattered recall for names that only ever appear in a directors table somewhere else in the file.

Two more checks, because a sample can't see everything

- **Manual precision audit over the whole document.** I went through all 635 replacements - 277 distinct mentions - by hand in `out/audit.jsonl`. No false positives.
- **Residual-PII scan.** I unzip the redacted .docx and re-scan every XML part with independent patterns for the PII I know is in the source. This catches leaks in places the detector never visited, and it did catch one.
- **Synthetic suite** (`--unit-tests`) for the four required types that don't appear in this document at all, plus negative controls.

Things worth being upfront about

- I wrote both the detector and the gold annotations, so precision of 1.000 should be read with that in mind. An independent annotator would be a much stronger test. The stratum B number is the more trustworthy signal - 3,183 tokens of ordinary prose sampled blind, no false positives.
- Recall is measured on a sample. The document-wide residual scan is my guard against a systematic blind spot, and it found one (bare locality names).
- These numbers are with NER off. Turning it on should help ORG recall and might introduce new false positives; I haven't been able to measure that run.

2. Results against the gold standard

6,968 tokens, 351 of them PII.

Metric	Value
Token accuracy	0.9986
Micro precision	1.0000
Micro recall	0.9715
Micro F1	0.9855
Macro precision	1.0000
Macro recall	0.9853
Macro F1	0.9922

Per type

PII type	Support	TP	FP	FN	Precision	Recall	F1
PERSON	83	83	0	0	1.000	1.000	1.000
ORG	85	75	0	10	1.000	0.882	0.938
ADDRESS	28	28	0	0	1.000	1.000	1.000
EMAIL	76	76	0	0	1.000	1.000	1.000
PHONE	50	50	0	0	1.000	1.000	1.000
URL	25	25	0	0	1.000	1.000	1.000
REG_ID	3	3	0	0	1.000	1.000	1.000
CIN	1	1	0	0	1.000	1.000	1.000

By stratum

Stratum	Tokens	Gold PII	Accuracy	Precision	Recall	F1
A - PII-dense	3,785	261	0.9989	1.000	0.9847	0.9923
B - random prose	3,183	90	0.9981	1.000	0.9333	0.9655

Every error that's left

All ten false-negative tokens are companies with no legal suffix. It's one failure mode, not ten:

Paragraph	Missed	Why
67	Financial Express, Jansatta, Loksatta	Newspaper names, no incorporation marker
97	Sterlite Copper	Brand name, written right after Vedanta Limited which was caught
122	Chakan Internal Kamgar Sangathna	Trade union; the KSH International prefix was caught, the rest wasn't

No false positives, no type confusions.

3. Synthetic suite

Covers the four required types that don't occur in this prospectus (SSN, credit card, date of birth, IP address), plus Aadhaar and PAN, plus negative controls.

Metric	Value
Token accuracy	1.0000
Precision / Recall / F1	1.000 / 1.000 / 1.000
Failing cases	0 of 8

The negative controls matter as much as the positive ones. None of these get touched:

- Order 411111111111112 shipped on ticket 100200300 for Fiscal 2025. - a 16-digit order number that fails Luhn, and a ticket number
- Refer to page 280 and section 12(1) of the Companies Act, 2013.
- The resolution passed on May 6, 2025 was filed with the Registrar of Companies. - a date with no birth cue anywhere near it

On the brief's precision question: I treat order and ticket numbers as not sensitive, and leave them intact.

4. The full document run

5,041 paragraphs scanned, 379 changed, 635 replacements across 277 distinct mentions, about 20 seconds.

PII type	Occurrences	Distinct mentions
PERSON	211	64
ORG	207	81
ADDRESS	74	53
EMAIL	52	26
PHONE	36	22
URL	26	14
REG_ID	13	6
CIN	9	4
DIN	7	7
Total	635	277

SSN, credit card, date of birth and IP address are all supported but genuinely don't occur here. I checked by scanning every XML part of the source package, not just the body text. The CLI says so explicitly rather than quietly printing a zero.

Residual scan of the output

Pattern	Distinct in source	Left in output
Email addresses	50	0
Phone numbers (+91)	20	0
CIN	4	0
SEBI registration numbers	6	0
Director Identification Numbers	7	0
Promoter and director surnames	10	0
Address localities	7	1

The one left is Birdewadi in Chakan Unit No. 2 (Birdewadi). It's a bare village name, which my scope rules say to leave alone.

That count of 50 distinct source emails includes ones that only exist inside hyperlink field codes and never appear as visible text. Before I handled `w:instrText`, 12 of them survived redaction while the visible text showed fakes. No amount of checking the body text would have caught that, which is the whole argument for scanning the output package instead of trusting the detector's own report.

5. Summary

	Precision	Recall	Accuracy
Gold standard, all types	1.000	0.972	0.9986
Gold standard, excluding suffix-less companies	1.000	1.000	1.000
Synthetic suite	1.000	1.000	1.000
Manual audit, 277 distinct mentions	1.000	-	-

I tuned this for precision. A redaction tool that mangles the document gets switched off and stops protecting anything, and under-redaction is something the audit log lets a human catch. The price of that choice is companies without an incorporation marker, and I've listed every one I know about.